

CURV AI — AWS Service Map

CURV AI Component	Current Technology	AWS Target	Required?	Cost Risk	Notes
API (FastAPI)	Python 3.12, uvicorn, port 8000	ECS Fargate behind ALB	Yes	Medium	Stateless; SSE needs ALB idle timeout $\geq 120s$; in-memory rate limiting won't scale >1 replica
Frontend (Next.js 15)	Node 20, port 3002, CSR SPA with API proxy	ECS Fargate behind ALB (NOT S3+CF)	Yes	Medium	Requires running Node server for <code>/api</code> rewrite proxy + 30-min <code>proxyTimeout</code> ; <code>output: "export"</code> not set
Celery Workers	Celery 5.4, Redis broker, 8 queue shards	ECS Fargate (separate service per queue group)	Yes	Medium	Worker image must include <code>prachar-shared</code> + <code>prachar-api</code> (loop.py imports API runtime)
Celery Beat	celery-beat scheduler	EventBridge Scheduler or singleton Fargate task	Yes	Low	Must never run >1 replica; weekly loop is the core product
PostgreSQL	PG 16, SQLAlchemy 2 async, 38 tables, RLS	RDS PostgreSQL 16 (db.t3.small)	Yes	High	Multi-AZ doubles cost; 11 tables missing RLS policies; no pgvector needed yet
Redis	Celery broker+backend, idempotency, progress, DLQ	ElastiCache Redis 7.1 (cache.t3.small)	Yes	High	2 nodes with failover = $\sim \$50$ /mo at t3.small; could use single-node for dev
Object Storage	MinIO (local), S3 (prod), reports + knowledge docs	S3 (1 bucket, KMS, lifecycle)	Yes	Low	$\sim \$2$ /mo for 20GB; currently only PDF reports use S3; video/images use fal.ai URLs
CDN / HTTPS	Render TLS (automatic)	CloudFront + ACM + Route53	Yes	Low	PriceClass_100; SPA fallback configured; OAI (should upgrade to OAC)
Load Balancer	Render proxy	ALB (internet-facing)	Yes	Medium	$\sim \$25$ /mo base + LCUs; HTTPS listener needs regional cert (not us-east-1)
NAT Gateway	N/A (Render)	1 NAT gateway	Yes	Medium	$\sim \$35$ /mo; SPOF if single-AZ; cross-AZ charges if multi-AZ
WAF	N/A	AWS WAF on ALB	Optional	Low	$\$5$ /mo + $\$1$ /rule; rate limit 2000/5min, SQLi, XSS rules
Secrets	<code>.env</code> file	Secrets Manager + Parameter Store	Yes	Low	$\sim \$0.40$ /secret/mo; DB creds, JWT, OAuth tokens, API keys
Container Registry	N/A	ECR (3 repos: api, web, worker)	Yes	Low	$\sim \$1$ /mo for stored images; lifecycle keep-10
Monitoring	stdout logs, <code>/metrics</code> (Prometheus), in-memory	CloudWatch Logs + Alarms + Dashboard	Yes	Low	5 alarms exist; SNS topic has NO subscription; no Sentry/OTel
CI/CD	GitHub Actions (ci.yml, deploy.yml)	GitHub Actions $\rightarrow$ OIDC $\rightarrow$ IAM role	Yes	Low	deploy.yml has broken Dockerfile paths; no OIDC (uses long-lived keys)
AI/LLM Providers	Groq, Anthropic, OpenAI, Gemini, fal.ai	External (no AWS service)	Yes	N/A	Keys in Secrets Manager; calls go to external APIs
Payments	Stripe, Razorpay	External (webhooks via ALB)	Yes	N/A	Webhook signature verification already implemented
Email	Resend + SMTP fallback	External (or Amazon SES)	Yes	N/A	RESEND_API_KEY in Secrets Manager; SES is cheaper alternative
Webhooks	Stripe, Razorpay, platform adapters	ALB $\rightarrow$ API (inline processing)	Yes	N/A	HMAC verification implemented; no SQS buffering

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SSE / Realtime	In-memory EventBus, StreamingResponse	ALB (sticky sessions or Redis pub/sub)	Yes	Low	Per-process EventBus breaks with >1 API replica; needs Redis pub/sub for multi-replica
Vector / Embeddings	In-memory VectorStore, JSONB embeddings	RDS pgvector (future)	Optional	Low	Currently uses hash pseudo-embeddings without OpenAI key; pgvector recommended post-launch
GPU Video Gen	fal.ai (Wan 3.0), external	External (no AWS GPU needed)	Optional	N/A	All video generation via fal.ai API; no GPU instances needed
PgBouncer	Configured in docker-compose	RDS Proxy or PgBouncer on ECS	Optional	Medium	Already PgBouncer-aware in code; RDS Proxy adds ~\$18/mo

Cost Summary by Tier

Tier	Monthly Cost	What's Included
\$25 budget	✗ Not achievable on AWS	Single EC2 VM with Docker (~\$15-30) or free-tier external services
Dev/Staging (minimal AWS)	~\$130-160/mo	1 Fargate task, db.t3.micro single-AZ, cache.t3.micro, ALB, NAT
Small Production	~\$275-310/mo	1 API + 1 worker, db.t3.small Multi-AZ, cache.t3.small x2, ALB, NAT, WAF, CloudFront
Growth Production	~\$1,220-1,280/mo	2 API + 2 worker, db.r6g.large Multi-AZ, cache.r6g.large x3, full WAF, CloudFront